

FAI seminar (weeks 1-2)

1 For lecture 1

Exercise 1 - PEAS exercise

For each of the following activities, give a PEAS description of the task environment and characterise it in terms of the properties listed in Lecture 1:

- Playing soccer.
- Exploring the subsurface oceans of Titan.
- Shopping for used AI books on the Internet.
- Playing a tennis match.
- Practicing tennis against a wall.
- Performing a high jump.
- Bidding on an item at an auction.

Exercise 2

Define in your own words the following terms: agent, agent function, agent program, rationality, autonomy, reflex agent, model-based agent, goal-based agent, utility-based agent, learning agent.

Exercise 3 - agent types

Consider a simple thermostat that turns on a furnace when the temperature is at least 3 degrees below the setting, and turns off a furnace when the temperature is at least 3 degrees above the setting. Is a thermostat an instance of a simple reflex agent, a model-based reflex agent, or a goal-based agent?

2 For lecture 2 (solving problems by searching)

Exercise 4

Define in your own words the following terms: state, state space, search tree, search node, goal, action, transition model, and branching factor.

Exercise 5

Your goal is to navigate a robot out of a maze. The robot starts in the center of the maze facing north. You can turn the robot to face north, east, south, or west. You can direct the robot to move forward a certain distance, although it will stop before hitting a wall.

1. Formulate this problem. How large is the state space?
2. In navigating a maze, the only place we need to turn is at the intersection of two or more corridors. Reformulate this problem using this observation. How large is the state space now?
3. From each point in the maze, we can move in any of the four directions until we reach a turning point, and this is the only action we need to do. Reformulate the problem using these actions. Do we need to keep track of the robot's orientation now?